

quantum computers are built with different architecture than standard computers, it's possible to design new algorithms that take advantage of their different physical properties to perform tasks with extra speed. For now, this is more or less only an academic concern, since quantum computers are not yet in a state where they are used for practical purposes. But if the technology ever matures, quantum algorithms could become extremely important.

When you learn about algorithms in these or many other fields, you will not be starting from scratch. By mastering the algorithms of this book, you've come to grasp they are, how they tend to function, and how to write code for them. Learning your first algorithm may have felt quite difficult, but learning your 50th or 200th will be much easier, since your brain will be used to the general patterns of how they are constructed and how to think about them.

To prove that you can now understand and code algorithms, we'll explore a few algorithms that work together to provide the functionality of a chatbot. If you can pick up how they work and how to write code for them in the short introduction provided here, then you're on your way to being able to pick up how any algorithm works in any field.

Building a Chatbot

Let's build a simple chatbot that can answer questions about the table of contents of this book. We'll start by importing modules that will be important later:

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from scipy import spatial
import numpy as np
import nltk, string
```

The next step we'll take to create our chatbot is *text normalization*, the process of converting natural language text to standardized substrings; it enables easy comparison between superficially different texts. We want our bot to understand that *America* and *america* refer to the same thing, that *regeneration* expresses the same idea as *regenerate* (albeit a different part